

SIMATS ENGINEERING

Assessment Tool 2 — Annexure A: Concept Mapping

Course name	CSA04 – Operating Systems
Course outcome	CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)
Assessment tool weightage	35%
Questions	5
Total marks	25
Student name	Kavinilavu S
Register number	192565038

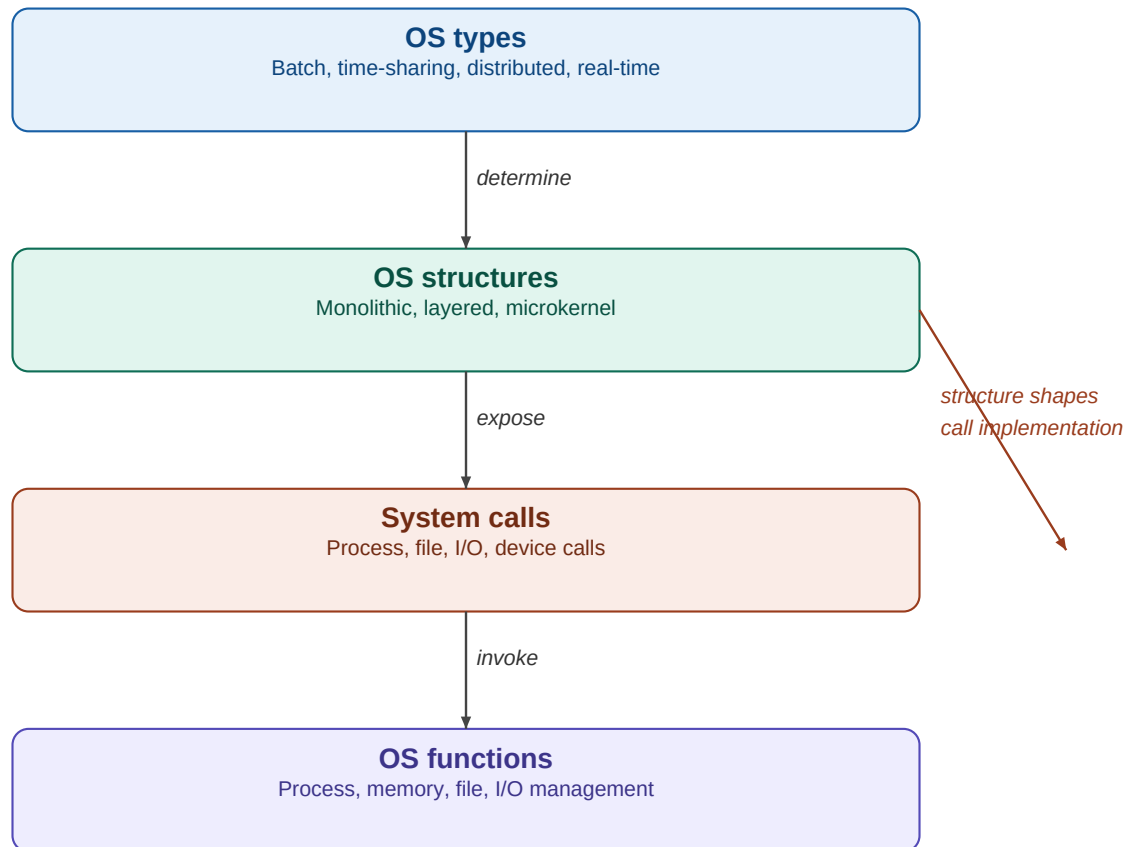
Code of conduct

I, Kavinilavu S (Reg. No. 192565038), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Question 1 (5 marks)

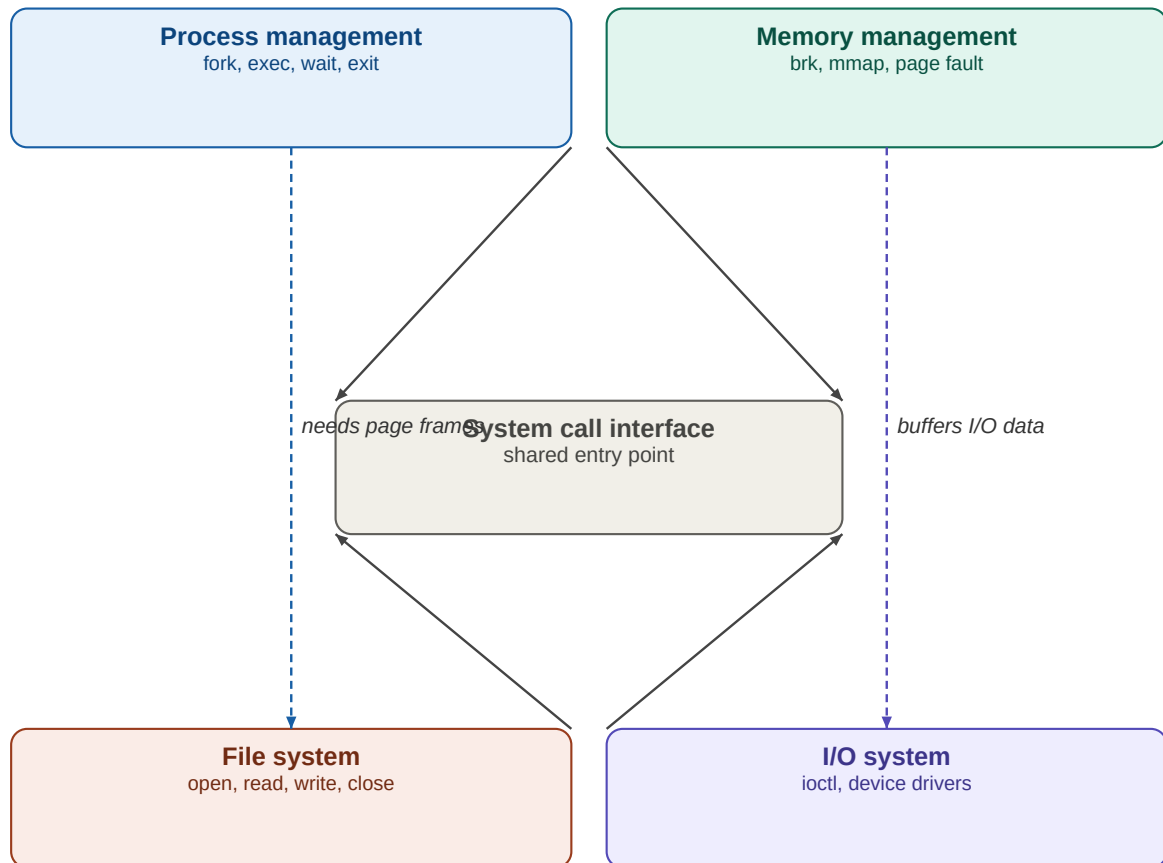
Construct a multi-level concept map integrating: Operating System Types → OS Structures → System Calls → OS Functions. Your map must show cross-links (e.g., how OS structure influences system call implementation).



The map flows top to bottom: OS type determines the structure chosen, the structure exposes a set of system calls, and those calls invoke the core OS functions. The coral cross-link on the right captures the key insight the question asks for: the OS structure directly shapes how a system call is implemented — a microkernel routes a call through message passing between servers, while a monolithic kernel services the same call with a single direct function invocation.

Question 2 (5 marks)

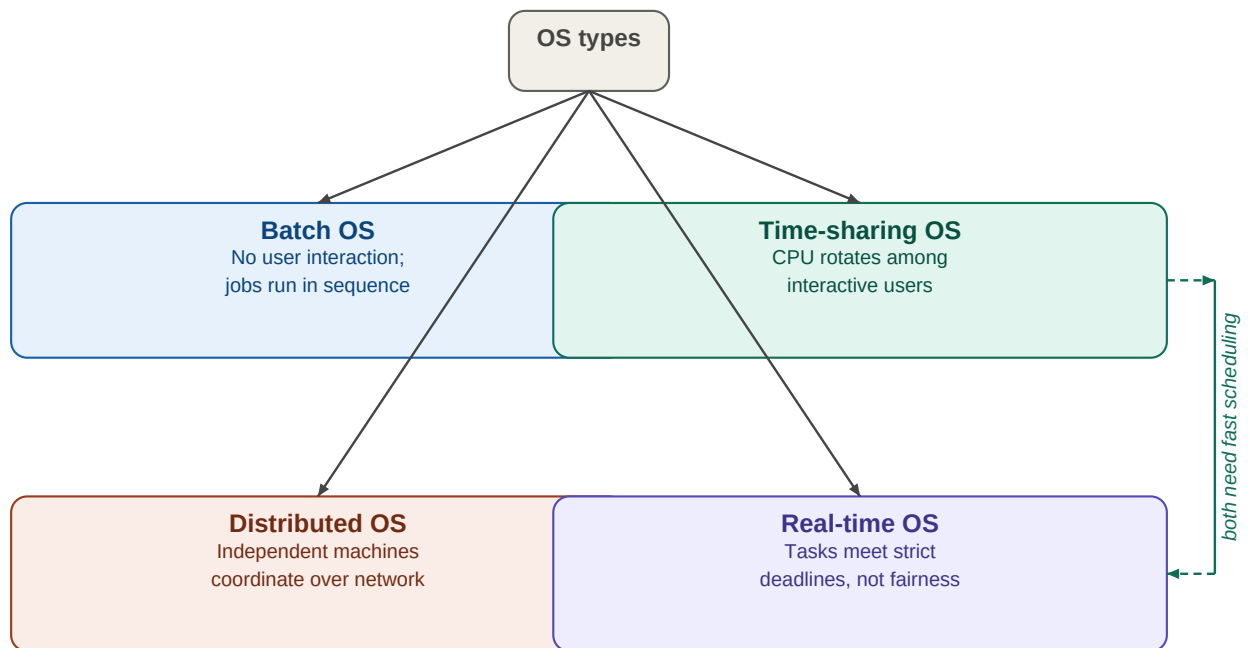
Design a comprehensive concept map representing: Process Management, Memory Management, File System, and I/O System. Include interdependencies and indicate where system calls interact with each component.



All four subsystems connect through the shared system call interface at the centre, which is how user programs reach each of them (fork/exec for process management, brk/mmap for memory management, open/read/write for the file system, and ioctl for the I/O system). The dashed side links mark the two direct interdependencies: process management needs memory management to allocate page frames for a new process, and the file system relies on the I/O system to actually move data to and from the physical device.

Question 3 (5 marks)

Create a comparative concept map for: Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS. Highlight differences, similarities, advantages, and limitations using linking phrases.

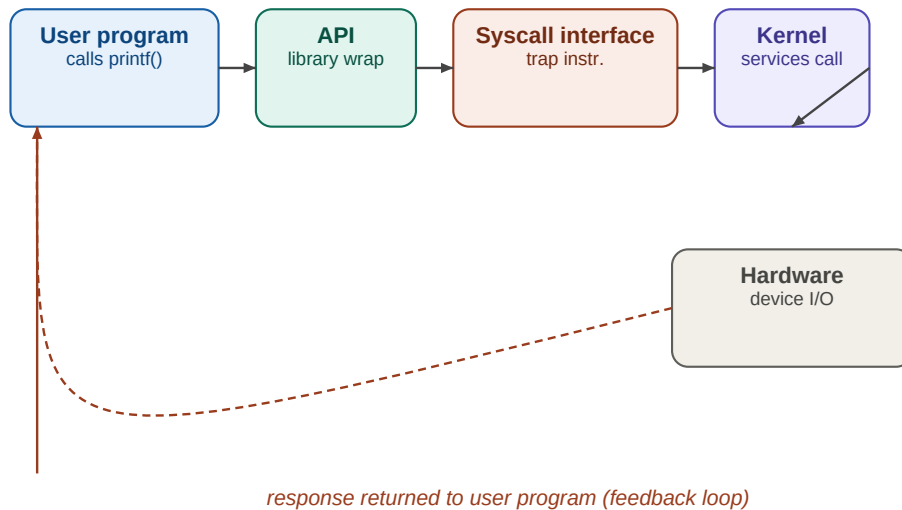


Shared limitation: batch and distributed OS both give the user no direct runtime control.

All four are types of operating system, shown branching from the root. Batch OS runs jobs sequentially with no user interaction; time-sharing OS rotates the CPU across many interactive users; distributed OS coordinates independent machines over a network; real-time OS guarantees strict task deadlines. The teal link on the right marks a similarity: time-sharing and real-time systems both depend on fast, responsive scheduling, even though their goals differ (fairness versus deadline guarantees). The note at the bottom highlights a shared limitation: batch and distributed systems both give the user little or no direct control at runtime.

Question 4 (5 marks)

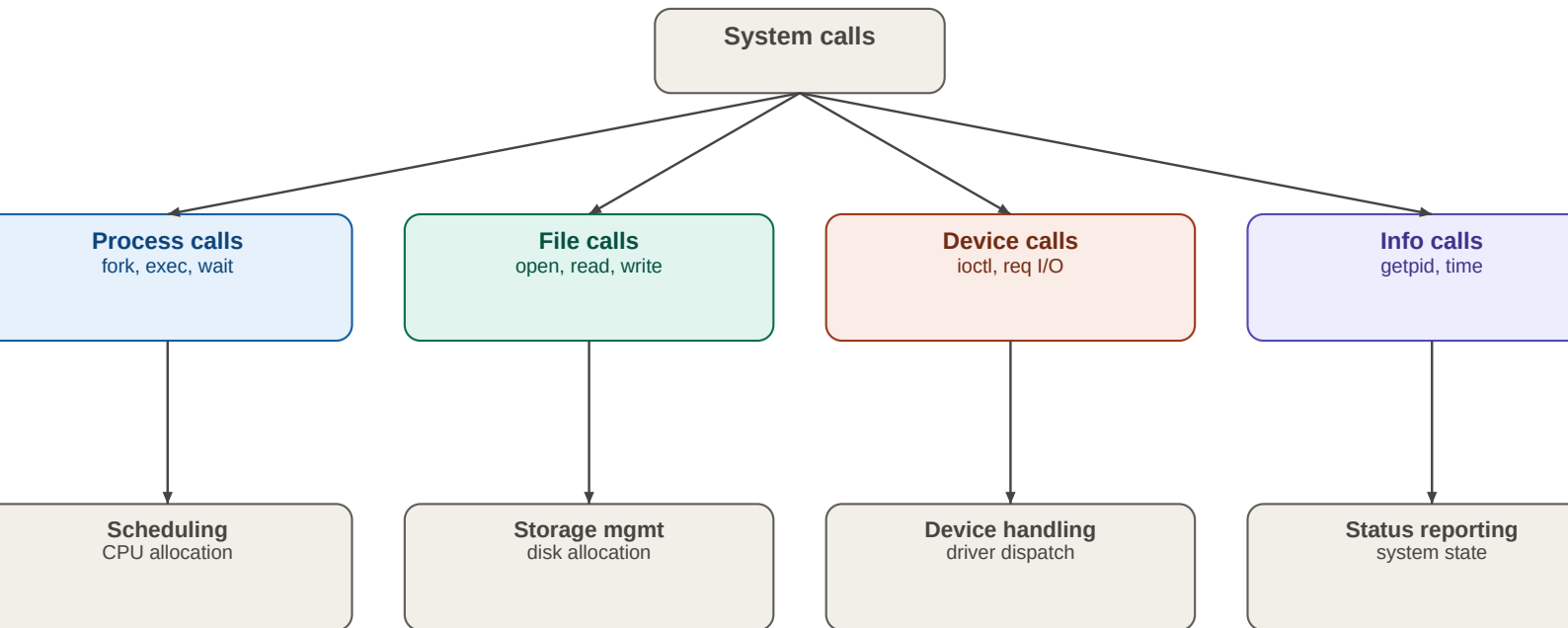
Develop a dynamic concept map showing the execution flow: User Program → API → System Call Interface → Kernel → Hardware → Response back to User. Include feedback loops and control flow paths.



The top row is the forward control path: a user program calls a library API (e.g. printf), which wraps the request and issues a trap instruction through the system call interface into the kernel; the kernel then services the request using the hardware. The dashed coral arrow is the feedback loop the question asks for: the result produced by the hardware is returned back up through the kernel and the API to the user program, completing the control-flow cycle.

Question 5 (5 marks)

Construct a hierarchical + functional concept map for system calls: Process, File, Device, and Information system calls. Also show how they relate to OS services like scheduling, memory allocation, and device handling.



System calls are grouped into four categories at the top level: process calls (fork, exec, wait), file calls (open, read, write), device calls (ioctl, request I/O), and information calls (getpid, time). Each category is then linked functionally, in the row below, to the OS service it ultimately relies on: process calls depend on the scheduler for CPU allocation, file calls depend on storage management, device calls depend on device handling, and information calls depend on the OS's status-reporting service.

Rubrics for calculation — Concept Mapping

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Clear, meaningful, and accurate relationships	Correct relationships with minor gaps	Weak, unclear, or partially incorrect relationships	No meaningful relationships established
Organization	20%	Well-structured hierarchically with logical flow	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Clear, neat, visually structured, easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks evaluation

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks
Q1						
Q2						
Q3						
Q4						
Q5						
Overall Total (100)						